

第九章 麦克斯韦方程和电磁波

$$\oint_S \vec{D} \cdot d\vec{S} = \int_V \rho_v \cdot dv$$

$$\nabla \cdot \vec{D} = \rho_v$$

$$\oint_L \vec{E} \cdot d\vec{l} = - \int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

$$\nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

$$\oint_S \vec{B} \cdot d\vec{S} = 0$$

($\Rightarrow$)

$$\nabla \cdot \vec{B} = 0$$

$$\oint_L \vec{H} \cdot d\vec{l} = \int_S (\vec{j} + \frac{\partial \vec{D}}{\partial t}) \cdot d\vec{S}$$

$$\nabla \times \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t}$$

介电: $\left\{ \begin{array}{l} \vec{D} = \epsilon_r \epsilon_0 \vec{E} \\ \vec{j} = \sigma (\vec{E} + \vec{v} \times \vec{B}) \\ \vec{B} = \mu_r \mu_0 \vec{H} \end{array} \right.$

1. 真空中各向同性

$$\left\{ \begin{array}{l} \nabla \cdot \vec{E} = 0 \\ \nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t} \\ \nabla \cdot \vec{H} = 0 \\ \nabla \times \vec{H} = \frac{\partial \vec{E}}{\partial t} \end{array} \right.$$

① $\vec{E}$ 的波动

$$\begin{aligned} \nabla \times (\nabla \times \vec{E}) &= \nabla (\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = - \nabla^2 \vec{E} \\ &= \nabla \times (- \frac{\partial \vec{B}}{\partial t}) = - \mu_r \mu_0 \frac{\partial}{\partial t} (\nabla \times \vec{H}) \\ &= - \mu_r \mu_0 \frac{\partial}{\partial t} \frac{\partial \vec{E}}{\partial t} \\ &= - \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} \end{aligned}$$

$$\Rightarrow \nabla^2 \vec{E} - \epsilon_r \mu \frac{\partial^2 \vec{E}}{\partial t^2} = 0 \quad c^2 = \frac{1}{\epsilon_r \mu} \quad c = \frac{1}{\sqrt{\epsilon_r \mu}}$$

② $\vec{H}$ 的波动

$$\begin{aligned} \nabla \times (\nabla \times \vec{H}) &= \nabla (\nabla \cdot \vec{H}) - \nabla^2 \vec{H} = - \nabla^2 \vec{H} \\ &= \nabla \times \frac{\partial \vec{D}}{\partial t} = \epsilon_r \epsilon_0 \frac{\partial}{\partial t} (\nabla \times \vec{E}) \\ &= - \epsilon_r \epsilon_0 \frac{\partial}{\partial t} (\frac{\partial \vec{B}}{\partial t}) \\ &= - \mu_r \mu_0 \epsilon_r \epsilon_0 \frac{\partial^2 \vec{H}}{\partial t^2} \end{aligned}$$

$$\nabla^2 \vec{H} - \epsilon_r \mu \frac{\partial^2 \vec{H}}{\partial t^2} = 0$$

$$\Rightarrow \begin{cases} \vec{E} = \vec{E}_0 \cos(\omega t - \vec{k} \cdot \vec{r}) \\ \vec{H} = \vec{H}_0 \cos(\omega t - \vec{k} \cdot \vec{r} + \varphi) \end{cases}$$

$$\begin{cases} \nabla \cdot \vec{E} = 0 & k \cdot \vec{E}_0 = 0 \\ \nabla \cdot \vec{H} = 0 & k \cdot \vec{H}_0 = 0 \end{cases} \Rightarrow$$

$$\nabla \times \vec{E} = -\mu_0 \mu_0 \frac{\partial \vec{H}}{\partial t}$$

$$\Rightarrow \begin{cases} \varphi = 0 \\ k \times \vec{E} = \mu_0 \mu_0 \omega \vec{H}_0 \Rightarrow \vec{k} \perp \vec{H}_0, \vec{E}_0 \end{cases}$$

2. 在均匀线性介质中, $D = \epsilon E$, $B = \mu H$

$$\text{设: } \vec{E}(r,t) = \vec{E}(r) \exp(-j\omega t)$$

$$\vec{B}(r,t) = B(r) \exp(-j\omega t) \quad \vec{H}(r,t) = H(r) \exp(-j\omega t)$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\mu \frac{\partial \vec{H}}{\partial t} \quad \nabla \cdot \vec{E} = 0$$

$$\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t} = \epsilon \frac{\partial \vec{E}}{\partial t} \quad \nabla \cdot \vec{B} = 0$$

代入:

$$\nabla \times \vec{E} = \mu j\omega H(r) \exp(-j\omega t)$$

$$\nabla \times \vec{H} = -\epsilon j\omega E(r) \exp(-j\omega t)$$

$$\begin{aligned} \nabla \times (\nabla \times \vec{E}) &= \nabla (\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\nabla^2 \vec{E} \\ &= \mu j\omega \nabla \times \vec{H} = -\mu \epsilon (j\omega)^2 \vec{E} \end{aligned}$$

$$\begin{aligned} \nabla \times (\nabla \times \vec{H}) &= \nabla (\nabla \cdot \vec{H}) - \nabla^2 \vec{H} = -\nabla^2 \vec{H} \\ &= -\epsilon j\omega \nabla \times \vec{E} = -\mu \epsilon (j\omega)^2 \vec{H} \end{aligned}$$

$$\Rightarrow \begin{cases} \nabla^2 \vec{E} + \mu \epsilon \omega^2 \vec{E} = 0 \\ \nabla^2 \vec{H} + \mu \epsilon \omega^2 \vec{H} = 0 \end{cases} \quad \text{解为 } \begin{cases} \vec{E} = \vec{E}_0 e^{i(\omega t - \vec{k} \cdot \vec{r})} \\ \vec{H} = \vec{H}_0 e^{i(\omega t - \vec{k} \cdot \vec{r})} \end{cases}$$

由此:

$$\hookrightarrow \varphi_E = \varphi_H, \quad \hookrightarrow \vec{k} \cdot \vec{E} = \vec{k} \cdot \vec{H}$$

$$W = \frac{1}{2} \epsilon_0 \epsilon_0 E^2 + \frac{1}{2} \mu_0 \mu_0 H^2$$

$$S = \frac{dW}{dt dV} = \frac{W dV}{dt dV} = W V = \frac{1}{\epsilon_0 \mu_0} \left(\frac{1}{2} (\epsilon_0 E^2 + \mu_0 H^2) \right)$$

$$\frac{E}{H} = \frac{\epsilon_0}{\mu_0} = \sqrt{\frac{\mu_0}{\epsilon_0}}$$

$$S = \frac{1}{2} \left(\frac{\epsilon_0}{\mu_0} E^2 + \mu_0 H^2 \right) = EH$$

$$\Rightarrow \vec{S} = \vec{E} \times \vec{H}$$

严格推导:

$$W = W_e + W_m = \frac{1}{2} \iiint_V (\vec{D} \cdot \vec{E} + \vec{B} \cdot \vec{H}) dV$$

$$\frac{dW}{dt} = \frac{1}{2} \iiint_V \frac{\partial}{\partial t} (\vec{D} \cdot \vec{E} + \vec{B} \cdot \vec{H}) dV$$

$$= \frac{1}{2} \iiint_V \left(\epsilon_0 \frac{\partial}{\partial t} (\vec{E} \cdot \vec{E}) + \mu_0 \frac{\partial}{\partial t} (\vec{H} \cdot \vec{H}) \right) dV$$

$$= \iiint_V \left(\epsilon_0 \vec{E} \cdot \frac{\partial \vec{E}}{\partial t} + \mu_0 \vec{H} \cdot \frac{\partial \vec{H}}{\partial t} \right) dV$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \quad \nabla \times \vec{H} = \vec{j}_0 + \frac{\partial \vec{D}}{\partial t}$$

$$= \iiint_V (\vec{E} \cdot (\nabla \times \vec{H} - \vec{j}_0) - \vec{H} \cdot (\nabla \times \vec{E})) dV$$

$$\nabla \cdot (\vec{E} \times \vec{H}) = \vec{H} \cdot (\nabla \times \vec{E}) - \vec{E} \cdot (\nabla \times \vec{H})$$

$$= \iiint_V (-\nabla \cdot (\vec{E} \times \vec{H}) + \vec{j}_0 \cdot \vec{E}) dV$$

$$= -\oint_S (\vec{E} \times \vec{H}) \cdot d\vec{s} - \iiint_V (\vec{j}_0 \cdot \vec{E}) dV$$

$$= -\oint_S \vec{S} \cdot d\vec{s} - \iiint_V p dV \quad p = \vec{j}_0 \cdot \vec{E} = \vec{j}_0 \cdot (\rho \vec{j}_0 - \vec{E})$$

$$= -\oint_S \vec{S} \cdot d\vec{s} - \underbrace{\int_0 R^2 + \int_0 z}_{\text{电源做功 - 焦耳热}}$$

能量在回路中传播:

通过空间中电磁场从导体的侧面输入

趋肤效应:

$$\nabla \cdot \vec{E} = 0, \quad \nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \quad \nabla \times \vec{H} = \vec{J}_0 = \sigma \vec{E}$$

$$\begin{aligned} \nabla \times (\nabla \times \vec{E}) &= \nabla (\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\nabla^2 \vec{E} \\ &= -\mu \frac{\partial}{\partial t} (\nabla \times \vec{H}) = -\mu \sigma \frac{\partial \vec{E}}{\partial t} \end{aligned}$$

$$\Rightarrow \nabla^2 \vec{E} - \mu \sigma \frac{\partial \vec{E}}{\partial t} = 0$$

$$\begin{aligned} \nabla \times (\nabla \times \vec{H}) &= \nabla (\nabla \cdot \vec{H}) - \nabla^2 \vec{H} = -\nabla^2 \vec{H} \quad (\vec{H}) \propto n \\ &= \sigma (\nabla \times \vec{E}) = -\sigma \mu \frac{\partial \vec{H}}{\partial t} \end{aligned}$$

$$\Rightarrow \nabla^2 \vec{H} - \mu \sigma \frac{\partial \vec{H}}{\partial t} = 0$$

$$\text{设 } \vec{E} = \vec{E}_0 e^{i(\omega t - k z)}$$

(B) η^L

$$\vec{H} = \vec{H}_0 e^{i(\omega t - k z)}$$

$$\Rightarrow \nabla^2 \vec{E} - \mu \sigma \frac{\partial \vec{E}}{\partial t} = \frac{\partial^2 \vec{E}}{\partial z^2} - \mu \sigma \frac{\partial \vec{E}}{\partial t}$$

$$= (-k^2) \vec{E} - \mu \sigma (i\omega) \vec{E} = (-k^2 - i\omega \mu \sigma) \vec{E} = 0$$

$$\text{解得 } k = \sqrt{-i\omega \mu \sigma} = \sqrt{\omega \mu \sigma} e^{i\frac{3\pi}{4}}$$

$$\text{代入, } \vec{E} = \vec{E}_0 e^{i\omega t} e^{-e^{i\frac{3\pi}{4}} k z}$$

$$= \vec{E}_0 e^{i\omega t} \cdot e^{-e^{i\frac{3\pi}{4}} \sqrt{\omega \mu \sigma} z}$$

$$= \vec{E}_0 e^{-\frac{\sqrt{2}}{2} \omega \mu \sigma z} e^{i(\omega t - \frac{\sqrt{2}}{2} \omega \mu \sigma z)}$$

衰减振幅.

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电磁波动量:

相对论情况下: $E = pc$

$$\hookrightarrow \text{动量密度 } \vec{g} = \frac{\vec{p}}{V} = \frac{\vec{E}}{v \cdot c} = \frac{1}{c} \vec{S} = \frac{\vec{S}}{c^2}$$
$$= \epsilon_0 \vec{E} \times \vec{H} = \vec{D} \times \vec{B}$$

角动量:

$$\vec{L} = \vec{r} \times \vec{g} = \vec{r} \times (\vec{D} \times \vec{B}) = \vec{D} (\vec{r} \cdot \vec{B}) - \vec{B} (\vec{r} \cdot \vec{D})$$

补充: 本题: 电磁场中圆形导线电流元1段电场的动

光压:

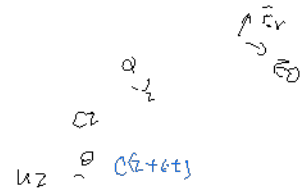
$$p = \frac{F}{S} = \frac{20P}{S_{\text{acc}}} = \frac{2gV}{S_{\text{acc}}} = 2gC = \frac{2S}{C} \quad g =$$

带电粒子的电磁辐射、

$t=0$ 静止, 在 $0 \sim t$ 时间内以加速度 a 运动, 此后以 $u=at$ 做匀速直线运动

在 $t+\tau$ 时:

$$\begin{aligned}\tan \alpha &= \frac{E_r}{E_\theta} \\ &= \frac{Qat - u \cos \theta}{u \tau \sin \theta} \quad (u \ll c) \\ &\approx \frac{Qat}{u \tau \sin \theta} = \frac{Q}{a \tau \sin \theta}\end{aligned}$$



$$\begin{aligned}E_\theta &= \frac{E_r}{\tan \alpha} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \cdot \frac{a \tau \sin \theta}{c} \\ &= \frac{Q a \sin \theta}{4\pi\epsilon_0 c^2 r}\end{aligned}$$

$$\begin{aligned}S &= \vec{E} \times \vec{H} = \left(\frac{\epsilon_0}{\mu_0} \right)^{1/2} E_\theta^2 = \frac{Q^2 a^2 \sin^2 \theta}{16\pi^2 \epsilon_0 c^4 r^2} \cdot \frac{1}{\epsilon_0 \mu_0} \\ &= \frac{Q^2 a^2 \sin^2 \theta}{16\pi^2 \epsilon_0 c^3 r^2}\end{aligned}$$

$$\begin{aligned}\frac{dW}{dt} &= \iint_{\text{surface}} S \cdot \hat{r} \sin \theta \, d\omega \, r^2 = r^2 \cdot \frac{Q^2 a^2}{16\pi^2 \epsilon_0 c^3 r^2} \int_0^{2\pi} d\phi \int_0^\pi \sin^3 \theta \, d\theta \\ &= \frac{Q^2 a^2}{8\pi\epsilon_0 c^3} \int_0^\pi \sin^3 \theta \, d\theta = \frac{Q^2 a^2}{6\pi\epsilon_0 c^3}\end{aligned}$$

振荡电偶极子的辐射电磁场

$$\vec{p} = p_0 \cos(\omega t) \hat{z}$$

①真空中匀速运动的带电粒子产生 $\vec{E}$, $\vec{B}$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^3} \vec{r} \rightarrow \frac{q}{r^3} \vec{r}$$

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{q \vec{v} \times \vec{r}}{r^3} \rightarrow \frac{1}{c} \frac{q \vec{v} \times \vec{r}}{r^3}$$

由上述讨论:

$$\text{加速在 } r \text{ 处产生 } E_r = \frac{q a}{4\pi\epsilon_0 r^2} \quad E_\theta = \frac{q a \sin\theta}{4\pi\epsilon_0 c^2 r}$$

$$\oint \vec{H} \cdot d\vec{L} = \iint_S \frac{\partial \vec{D}}{\partial t} \cdot d\vec{S} \quad \Rightarrow \quad \frac{\partial}{\partial t} \iint_S \vec{D} \cdot d\vec{S} = \frac{\partial \phi_D}{\partial t}$$

或 $\oint \vec{E} \cdot d\vec{L} = -\frac{\partial \phi_D}{\partial t}$

$$\rightarrow V(-\frac{z}{r})$$

$$\phi_D = \epsilon_0 E_r \cdot S$$

$$= \epsilon_0 \frac{q}{4\pi\epsilon_0 r^2} \cdot 2\pi r^2 (1 - \cos\theta)$$

$$= \frac{q}{2} (1 - \cos\theta)$$

$$\frac{\partial \phi_D}{\partial t} = \frac{q}{2} \sin\theta \cdot \frac{d\theta}{dt}$$

$$q \frac{d}{dt} \frac{1}{r}$$

$$\frac{d}{dt} \left(\frac{1}{r} \right) = \frac{1}{r^2} \frac{dr}{dt} = \frac{1}{r^2} \frac{dr}{d\theta} \frac{d\theta}{dt} = \frac{1}{r^2} \frac{dr}{d\theta} \frac{d\theta}{dt}$$

$\rightarrow H_\varphi$

$$r a_\theta \approx \omega \cdot r \sin\theta \rightarrow a_\theta \approx \frac{\omega^2 r \sin\theta}{r} \approx \frac{\omega^2 r \sin\theta}{c}$$

$$\frac{\partial \phi_D}{\partial t} = \frac{q}{2} \sin\theta \frac{a_\theta}{c} \cdot a = \frac{q a \sin^2\theta}{2c}$$

$$H_\varphi = \frac{1}{2\pi r \sin\theta} \cdot \frac{\partial \phi_D}{\partial t} = \frac{q a \sin\theta}{4\pi c r} \quad B_\varphi = \frac{\mu_0 q a \sin\theta}{4\pi c r}$$

电磁单位制:

(1) 静电场

$$\text{库仑力: } F_{12} = k_e \frac{q_1 q_2}{r_{12}^2} \rightarrow \begin{cases} F_{12} = q_2 q_1 E \\ E = \frac{k_e}{q_2} \cdot \frac{q_1}{r_{12}^2} \end{cases} \quad \text{选取 } q_2 = 1$$